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Advanced Geomechanical and Natural Fracture Characterization Modelling for Production Enhancement in the Mauddud Carbonate Formation, Greater Burgan, Kuwait

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Abstract

The primary objective of this paper is to introduce fit-for-purpose technologies and best practices aimed at increasing the productivity of the Mauddud formation in the Greater Burgan Field, Kuwait, through detailed geomechanical modelling, natural fracture prediction, and a dual-porosity / dual-permeability reservoir model. Objectives include constructing a discrete fracture network and a 3D geomechanical model to better understand reservoir fluid flow and select sweet spots for new horizontal infill wells. The 3D geomechanical model provided the foundation for stress modelling and fracture design. It incorporates natural fractures based on numerical fracture prediction techniques, with faults and fractures embedded into a 25 million grid cell geomechanical grid. A new compressional/shear slowness correlation was established and served as the input for Young's Modulus, Poisson's Ratio, uniaxial compressive strength, tensile strength and friction angle modelling. These results were derived by a mix of co-kriged properties, machine learning from extensive laboratory and field data. Blind test models cross-validated against field data proved the model calibration, though the friction angle correlation was limited by data availability. The vertical stress gradient averages at ~0.98 psi/ft; pore pressure ranges from 0.44-0.48 psi/ft. The maximum horizontal stress is oriented ~40-50° from North. The modelling of fracture height is mainly controlled by local stress magnitudes and the surrounding shale formation, with advanced fracture modelling separating the different types of fractures into fractures related to shear and joint tectonic activity; fractures around the fault damage zone; and fractures derived from seismic attributes to capture the heterogeneity and complexity of the field. It was discovered that pressure communication between the two reservoirs, BGSU and Wara, is likely. Main recovery is driven by natural fractures, low matrix permeability, and pressure depletion. Based on the geomechanical model, optimal well placements were suggested for more than 100 planned wells.

Introduction

This study assesses and models the thin carbonate Mauddud reservoir in Kuwait's Greater Burgan field using a 3D geomechanical model integrated with natural fracture prediction (DFN) and seismic data. The goal is to boost field production by 2030. Analyses based on the model included reservoir integrity, natural fracture modelling, prediction of geomechanical properties, the stability of existing faults and fractures within the formation and served as the main input for hydraulic fracturing assessment.

Geological Background

The Greater Burgan Field, situated in southeastern Kuwait, ranks among the largest hydrocarbon accumulations worldwide, comprising a series of superimposed reservoirs spanning stratigraphic intervals from the Lower Cretaceous to the Tertiary. Within this complex petroleum system, the Mauddud Formation constitutes a Lower Cretaceous carbonate succession deposited in a shallow-marine ramp setting. The formation is typified by laterally extensive, thin-bedded limestones and dolomitic intervals intercalated with marls and argillaceous layers, which act as intraformational flow barriers.

The Mauddud carbonate matrix exhibits inherently low permeability typically in the order of tens of millidarcies or less paired with moderate porosity, averaging 15–20%. Reservoir quality is markedly influenced by diagenetic overprints, including cementation, dissolution, and dolomitization, resulting in pronounced heterogeneity across multiple scales. Natural fractures represent a key permeability enhancement mechanism, particularly in proximity to tectonic lineaments (Al-Bazzaz et al., 2006). Structurally, the Mauddud Formation is dissected by a network of NE–SW-oriented normal faults, emplaced during regional tectonic events in the Late Cretaceous and Tertiary (Alzankawi et al., 2014).

Integrated reservoir characterization, leveraging core, wireline log, seismic, and production datasets, has supported explain the spatial distribution of facies and petrophysical properties in many studies (e.g., Haas et al, 2022). Depositional facies range from mid-ramp peloidal grainstones to back-ramp mudstones, with the highest reservoir quality associated with high-energy shoal facies. However, lateral and vertical connectivity remains constrained in the absence of open, transmissive fracture networks. Static reservoir modelling has delineated multiple discrete flow units within the Mauddud, each characterized by distinct porosity-permeability relationships and capillary pressure responses. The interplay between these flow units and the fracture framework exerts primary control on well productivity. Volumetric assessments underscore the Mauddud reservoir's strategic significance to the Greater Burgan Field's hydrocarbon portfolio, despite the technical complexities inherent to its development.

Methodology

A multidisciplinary approach (Haas et al, 2023) was applied to characterize the Mauddud Formation, combining geomechanics with natural fracture prediction, and reservoir modelling. Both 1D and 3D mechanical earth models (MEMs) were constructed to simulate subsurface stress conditions and evaluate fault, discrete (natural) fracture networks (DFN) and fracturing potential (Haas et al. 2024). Machine learning algorithms aided in scenario evaluation to optimize field development. The integration of seismic, petrophysical, and geological inputs ensured robust model calibration, enabling scenario testing to identify sweet spots and maximize oil recovery across the Greater Burgan field.

The fundamental approach for any geomechanical analysis is to integrate all available data of rock properties, deformability, in-situ stresses, and pore pressure into a mechanical earth model. The MEM is then calibrated using all available measurements and observations to arrive at a consistent subsurface representation of the key geomechanical properties and parameters needed for subsequent analyses and engineering designs. The MEM includes time-dependent components that reflect geomechanical changes occurring over the life of a field such as pore pressure changes with production and injection operations

and can be updated at any point in future modelling processes. The identification of geomechanics related drilling events encountered in existing wells provided good calibration and validation to accurately determine the in-situ stresses and constrain the models. To create synthetic shear slowness log, multi-well cross plots were generated using the wells in which both shear and compressional logs were available. Three various correlations were developed based on the lithology. A blind test was performed on one of the wells in which shear slowness data was acquired from 3D far-field sonic (3DFFS) logs and showed a reasonable match.

The stability of rock under different loading conditions in the subsurface environment during drilling and production is controlled by the elastic, strength and stress state of the formation. The rock mechanical properties include elastic constants such as Young's modulus (YME), Biot's coefficient, and Poisson's ratio (PR), and rock strength data such as unconfined compressive strength (UCS), internal friction angle (FANG) and tensile strength (TSTR). Log data, notably compressional and shear slowness and rock bulk density, were utilized to compute the dynamic rock mechanical properties along depth in each formation interval. The static rock properties profiles were estimated using correlations (Figure 1) obtained from laboratory test, local field base and the literature (e.g., Plumb, 1994). Laboratory core test data including multi-stage triaxial tests conducted on Mauddud reservoir samples from nearby wells were provided based on which the rock strength and elastic properties were calculated.

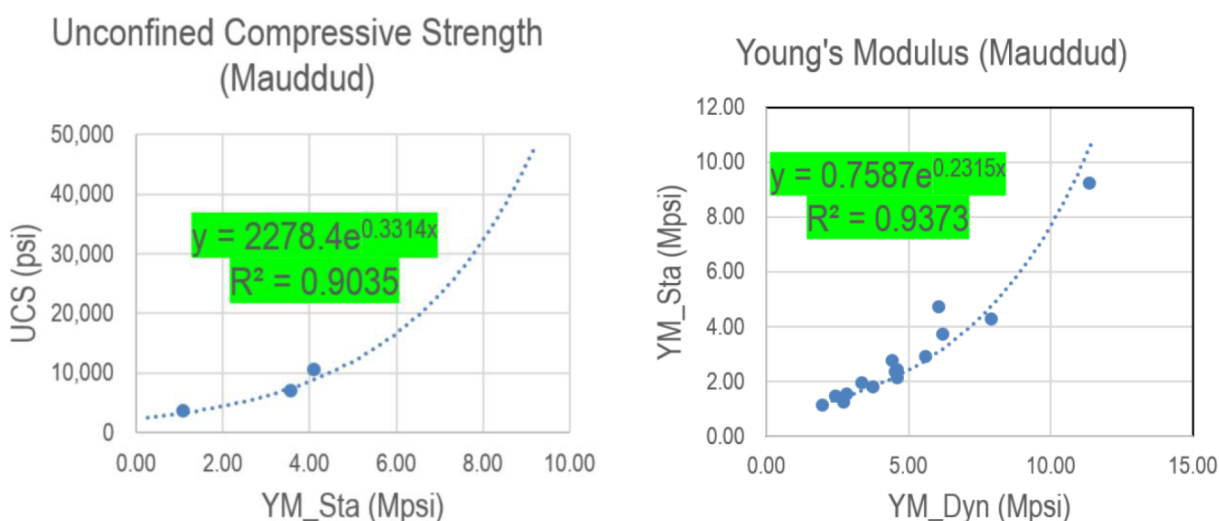


Figure 1—Correlations used for the calculation of elastic (dynamic to static) and rock strength parameters, exemplarily for the UCS (left) and YME (right).

Geological features such as discontinuities (faults, fractures) and varying structural geometry influence the stress state within a reservoir. To capture this varying behaviour on field-scale, 3D geomechanics modelling is required, which provides the numerical solution of stress states in each grid cell defined for intact, faulted and fractured rock. The numerical stress solutions allow the assessment of wellbore stability analysis including mud weight cube predictions and hydraulic fracture assessments to calibrate the model based on previous 1D MEM results. Data on pore pressure across the reservoir, any indication of the in-situ stress, and laboratory tests to calibrate strength and elastic rock properties are among essential model inputs. Once calibrated with field and/or borehole data, the model serves as a basis for analysing and interpreting the post-processed results. The results are investigated in terms of macroscopic observables such as mud weight predictions (MWP) and discontinuity stability analysis (DSA). Figure 2 illustrates the selected 3D MEM approach with respect to wellbore stability analyses (WBS), mud weight predictions (MWP) and the main input for further stimulation designs.

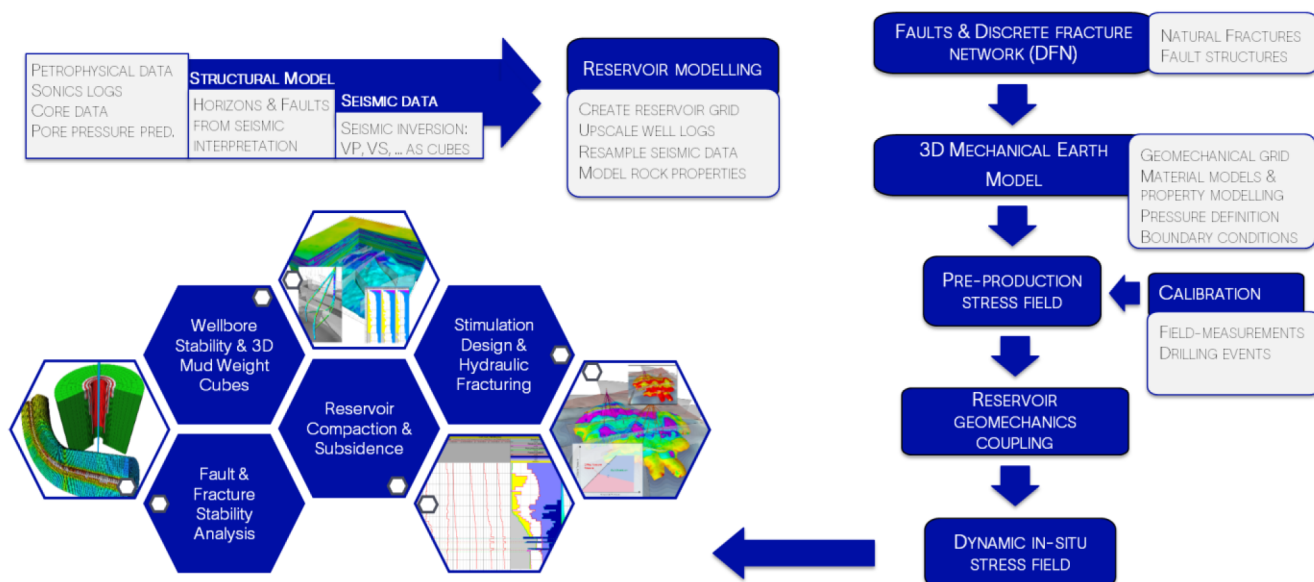


Figure 2—3D geomechanics workflow including all available data such as reservoir engineering, geology, seismic and faults/fractures (DFN). These data culminate in the analysis for WBS and stimulation designs (Haas et al. 2024).

Seismic data were used (Herwanger and Koutsabeloulis, 2011) for the calculation and distribution of geomechanical parameters (Figure 3) based on the 1D MEM results and co-kriged with the interval velocity (V_p) as well as the acoustic impedance (AI). In addition to the field-scale quality control and assessment, four (4) wells were selected with varying Mauddud thickness to verify the different log and seismic responses for the acoustic impedance and interval velocity in the Mauddud formation.

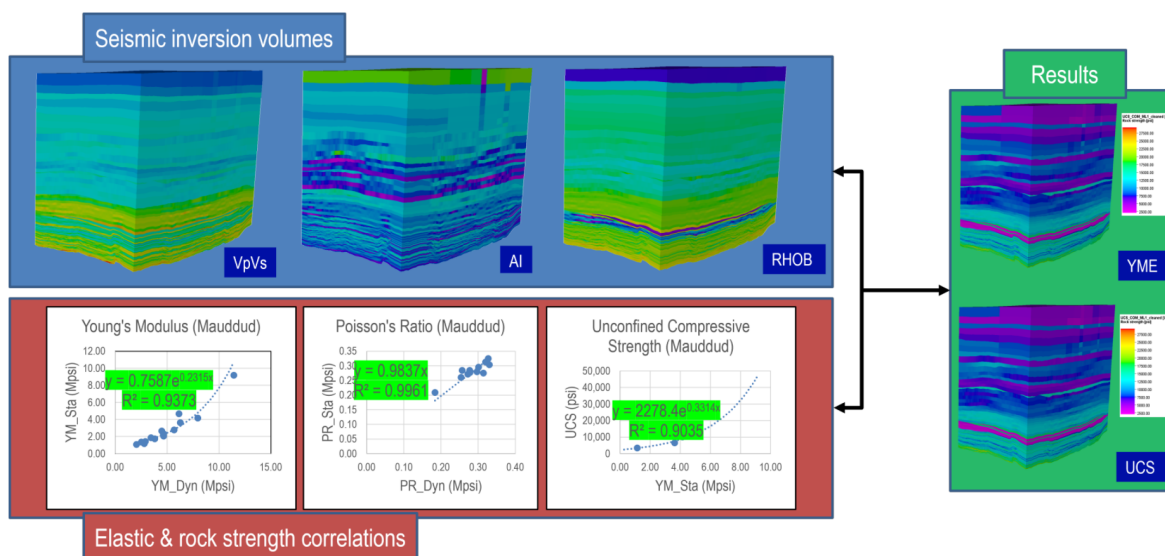


Figure 3—Workflow approach using the seismic inversion data (VpVs, AI, RHOB) to directly calculate the geomechanical properties based on rock correlations.

The construction of the 3D geomechanical grid included the process of embedding a high-resolution structural grid by adding an overburden, sideburden and underburden, respectively. The geomechanical properties were modelled based on 1D MEM logs, seismic and guided machine learning algorithms (Figure 4).

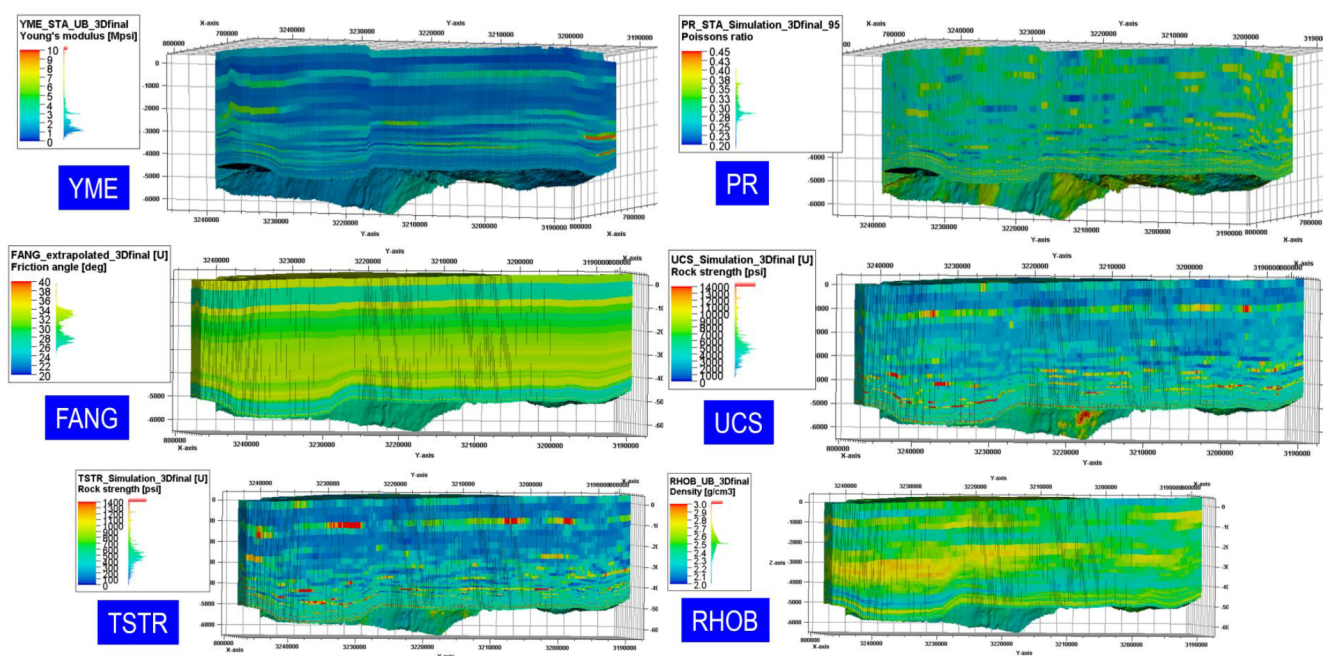


Figure 4—Modelled YME, PR, UCS, TSTR, FANG and RHOB across the full Greater Burgan field.

To account for the behaviour of faults, more than 430 faults were incorporated into the geomechanical grid. The normal and shear stiffnesses of faults represented weakened grid cells compared to the surrounding intact rock. These stiffnesses were taken for both faults and fractures, whereas for faults a friction angle of 23° was taken, while for fractures a value of 21° was chosen based on field data, field experience and fracture behaviour in carbonate formations based on the literature. The initial tectonic load was adapted iteratively for each performed numerical simulation. Based on field observations, the direction of minimum horizontal stress was set to $N130^\circ$ (measured clockwise from North), with minimum and maximum horizontal strains of 0.00005 and 0.0003, respectively. For the natural fracture prediction, borehole image data (BHI) and 3D far-field sonic (3DFFS) data were analysed and incorporated into the geomechanical model. This led to a sophisticated, robust discrete fracture network model through the Greater Burgan field.

Results and Discussion

Borehole breakouts, drilling-induced tensile fractures (DITFs), and borehole image logs were used to constrain the present-day stress regime for both the minimum and maximum horizontal stresses (Figure 5). The results show that the Mauddud reservoir is characterized by a strike-slip to normal faulting regime, with a maximum horizontal stress (SHmax) oriented approximately NE-SW, parallel to the main fracture and fault trends identified in seismic interpretation. No major drilling events across the Mauddud formation, tight spots or partial seepage losses (across Burgan) were identified. In the analysed wells, the Mauddud formation was drilled either in the 8-1/2" or 12-1/2" section with a drilling mud weight of 9.2~11.2 ppg. No significant over-pressure was noticed in the zones of interest. Also, in most of the wells there is no significant pore pressure change across Mauddud formation. The average pore pressure gradient across Mauddud for the selected wells was in the range of 0.44 – 0.48 psi/ft. The pore pressure was calibrated using valid field data in the selected wells. Underlying or overlying shale layers are assumed to be at hydrostatic pressure with no additional data available.

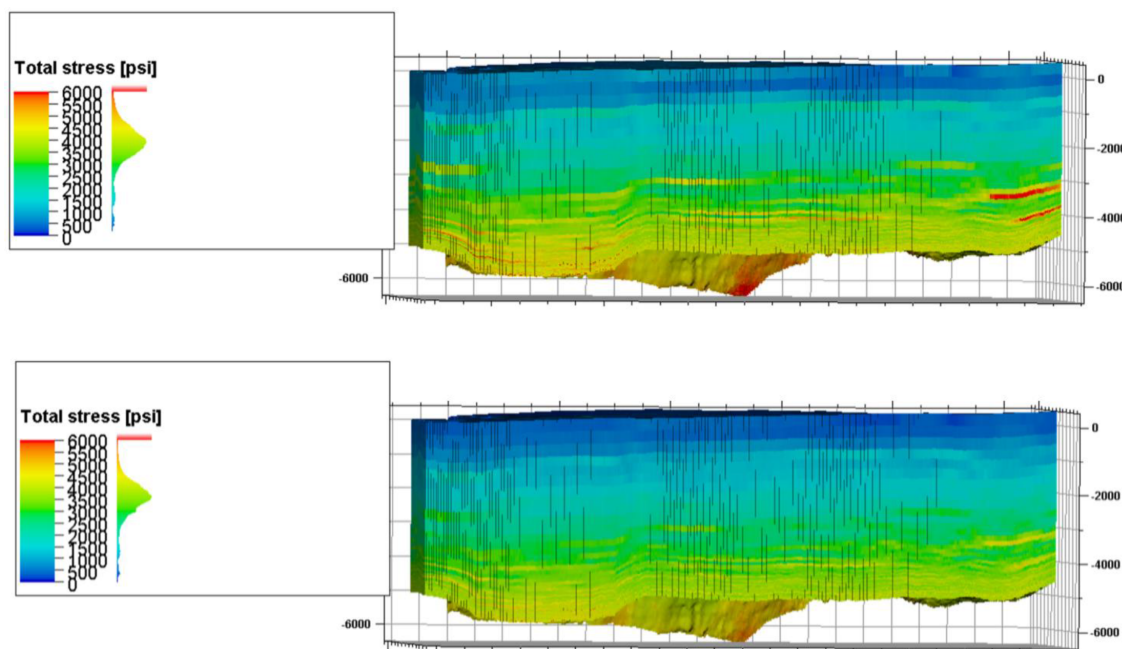


Figure 5—Maximum (top) and minimum (bottom) horizontal stress simulation results throughout Greater Burgan.

For the wellbore stability window (Figure 6), the main curves are related to pore pressure and to wellbore collapse. The breakout curve represents the most conservative limit related to wellbore instability, which means that if the mud weight (MW) goes below this limit, breakout or rock failure will occur. Pore pressure (PP) behaves as such that if $MW < PP$, there is risk of a kick in permeable rock. The fracture gradient (FG), or loss gradient (LG), is the minimum in-situ stress, which was also used for calibration with the closure pressure from field data.

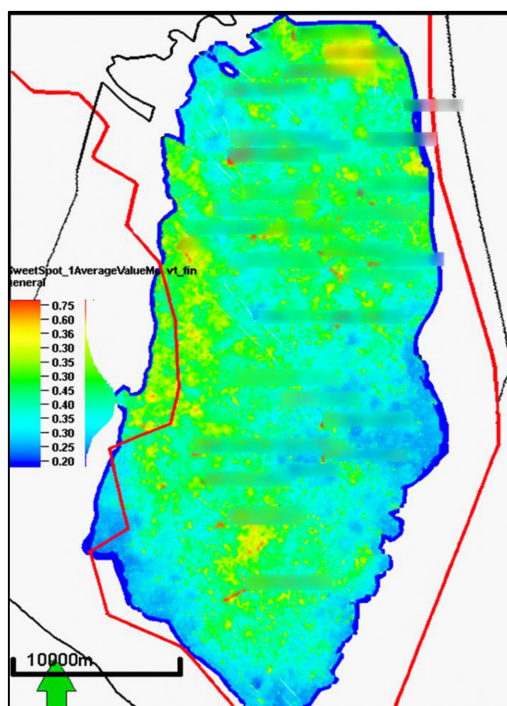


Figure 6—Sweet spot analyses for optimal well placements.

The fractures in Mauddud formation are oriented NE-SW, NW-SE and N-S direction, with the N-S component being more dominant in the Ahmadi area. Most of the fractures are high angle (>30 degrees). They appear to be fault related representing wall damage and localized deformation, formed as result of multiple tectonic episodes. Three fracture drivers, the fault damage zone (FDZ), fault natural fracture prediction (Fault-NFP) and seismic extracted patches (SE) were modelled and captured the heterogeneity in orientation and intensity of fractures across the field. This led to six fracture sets in total, covering the shear fractures, joints and fault damage zone. Fracture apertures are distributed stochastically based on fracture aperture distribution from borehole image analysis. Length of fractures appear to have significant impact on upscaled properties e.g., fracture permeability, and it should be optimized further in future work. When considering surface well-head locations of all previous wells drilled in Greater Burgan and sweet spot zones across the acreage this would require around 94 future horizontal Mauddud wells to drill up the acreage. Under this scenario the total Mauddud recovery factor is likely to be around 11%. One of the main benefits of performing a calibrated reservoir simulation history match is to identify sweet spots across the acreage for future development. The following "calibrated" properties were made available from the final full-field history matched model: reservoir pressure, intact rock permeability, fracture permeability, flow capacity, oil saturation, and the minimum horizontal stress. The goal was to identify planned well spots that would deem the most favourable. Therefore, a sweet spot analysis indicator (Figure 7) was introduced, which incorporated all five parameters. Hence, this implied to target areas with the lowest minimum horizontal stress to facilitate hydraulic fracturing. For the permeabilities and oil saturation, high values were targeted since they increase the productivity of the formation.

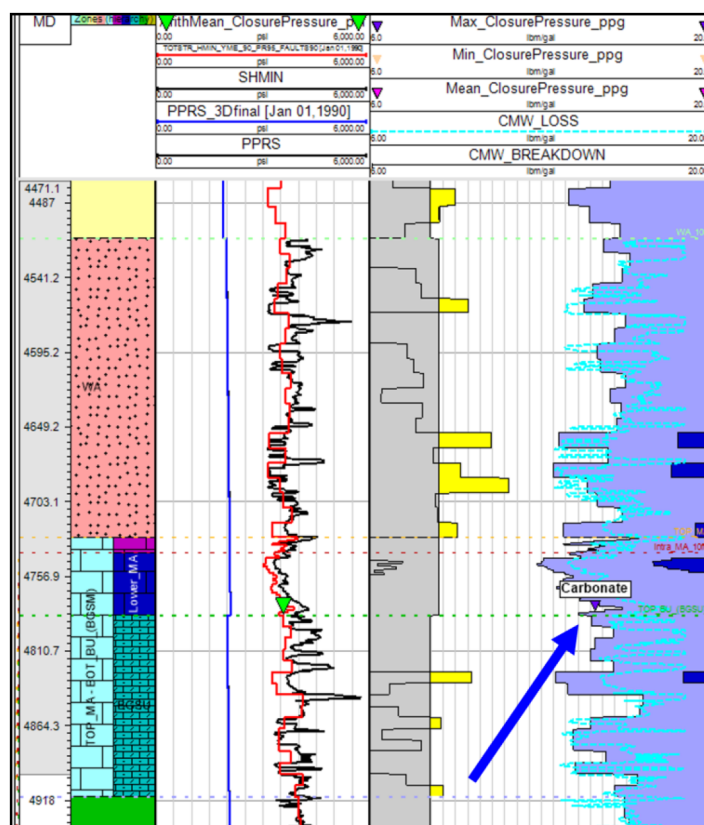


Figure 7—Wellbore stability analysis for one of the wells in Greater Burgan, comparing the 3D results (coloured area curves) versus the 1D results as single, black curves (left) and single colour (right). Note the good match (blue arrow) with calibration data (closure pressure).

For Mauddud's future well locations, two locations have been proposed in the north dominated by likely higher reservoir pressure (being away from intense drilling activity), larger Mauddud thickness as well as stable drilling mud windows. Another two locations have been proposed in the northern-mid region dominated by well-opened, natural fractures and are slightly away from the main reservoir pressure sink due to production from wells.

Conclusion

This integrated study enhances the understanding of the Mauddud carbonate reservoir's geomechanical behaviour in Greater Burgan. By fusing 1D/3D geomechanical modelling with natural fracture prediction and dual-porosity / dual-permeability reservoir engineering, the results have reshaped the impact on development strategy for the Mauddud Formation, focusing on prioritizing horizontal drilling in fracture-enhanced corridors through sweet spot analyses, applying multi-stage acid fracturing (see ADIPEC 2025 paper SPE-229962-MS), and deriving a solid fracture model. This study offers new insights into strike-slip and normal stress regimes, fracture systems, and production optimization strategies. The geomechanics-driven natural fracture prediction served as one of the main inputs together with the 3D geomechanical model and the dual-porosity / dual-permeability model, to identify optimal planned well locations and forecast increased production rates. Furthermore, the wellbore stability analyses guide the drilling process of new wells based on a better understanding of stress. It is highly recommended to place the future horizontal wells within the vicinity of the faults and opening fractures. Well placement should be focused on the northern to mid-northern regions of the Greater Burgan field. The insights of reservoir geomechanics and natural fracture prediction strongly support strategic decision-making, recovery improvement and production maximization. The approach marks a step forward in rapid, data-driven reservoir characterization and development planning, providing a replicable model for similar carbonate systems in Kuwait and globally.

Acknowledgments

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References

1. Al-Bazzaz, A. Gupta. Reservoir Characterization and Data Integration of Complex Mauddud-Burgan Carbonate Reservoir, SPE Europec/EAGE Annual Conference and Exhibition, June 12–15, 2006. SPE-100318-MS.
2. Alzankawi, O., Al-Houti, R., Ma, E., Ali, F., Alessandroni, M., Alvis, M. 2014. Mauddud Fractured Reservoir Analysis, Greater Burgan field: Integrated Fracture Characterization using Static and Dynamic Data. International Petroleum Technology Conference, Doha, Qatar, January 2014. IPTC-17471-MS.
3. Haas, M., Carraro, D., Ventra, D., Plötze, M., De Haller, A., Moscariello, A. 2022. Integrated stratigraphic, sedimentological and petrographical evaluation for CERN's Future Circular Collider subsurface infrastructure (Geneva Basin, Switzerland-France). *Swiss Journal of Geosciences* **115**, 16.
4. Haas, M., Al-Hamad, N., Heiland, J., Hussein, A. M., Shekhar, S., Srivastava, D. N., Ramon, J., Abdel-Basset, M., Al-Anzi, A. 2023. 4D Geomechanics New Modelling Approach Using Machine Learning for Drilling Event Predictions in the Giant North Kuwait Jurassic Field, SPE-216266-MS, Abu Dhabi, ADIPEC 2023.
5. Haas, M., Cifuentes, H. R., Ahumada, J. S., Suarez, Y. O., Acosta Gomez, J. C., Abad, F., Sorgi, C., Hussein, A. M. 2024. Drilling Integrity Analysis and Optimal Well Placement Based on

- 4D Coupled Geomechanical Modelling and Natural Fracture Prediction, SPE-217948-MS, Galveston, Texas, IADC 2024.
6. Herwanger, J., Koutsabeloulis, N. 2011. Seismic Geomechanics: How to Build and Calibrate Geomechanical Models using 3D and 4D Seismic Data. *EAGE Volume 5 of Education Tour Series*.
 7. Plumb, R.A. 1994. Influence of composition and texture on the failure properties of clastic rocks. EUROCK 94, Rock Mechanics in Petroleum Engineering Conference, Delft, The Netherlands, 13–20 pp.